Comparative evaluation of apical sealing ability and post-endodontic pain using bioceramic sealers versus epoxy resin sealers: a prospective randomized clinical trial

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Abstract. This prospective randomized clinical trial aimed to compare the post-endodontic pain and periapical healing outcomes, serving as a clinical indicator of apical sealing ability, of bioceramic sealers versus traditional epoxy resin sealers in the root canal treatment of teeth with necrotic pulps. A total of 120 patients diagnosed with asymptomatic apical periodontitis were randomly allocated into two groups ($n = 60$ per group). Group 1 received root canal obturation using a bioceramic sealer (BioRoot RCS) combined with a single-cone technique, while Group 2 was treated with an epoxy resin sealer (AH Plus) using warm vertical compaction. Postoperative pain was assessed by patients using a 100-mm Visual Analogue Scale (VAS) at 24 hours, 48 hours, and 7 days. Periapical healing, reflecting the long-term apical sealing ability and biological compatibility of the materials, was evaluated at the 12-month follow-up using the Periapical Index (PAI) score on standardized digital periapical radiographs. The bioceramic sealer group exhibited significantly lower mean VAS pain scores at 24 hours (18.4 ± 12.3 mm) and 48 hours (8.2 ± 7.5 mm) compared to the epoxy resin group (32.6 ± 15.1 mm and 16.4 ± 10.2 mm, respectively; $p < 0.001$). By day 7, pain levels were negligible and statistically comparable in both groups ($p = 0.412$). At the 12-month radiographic evaluation, the bioceramic group demonstrated a significantly higher rate of complete periapical healing (PAI score 1; 94.8%) compared to the epoxy resin group (86.4%; $p = 0.038$), indicating superior long-term apical sealing and tissue integration. In conclusion, bioceramic sealers significantly reduce early postoperative pain and promote superior periapical healing compared to conventional epoxy resin sealers, strongly supporting their clinical efficacy, biocompatibility, and excellent sealing properties as a highly reliable alternative for modern root canal obturation.

1 Introduction

The primary objective of root canal treatment is to prevent or treat apical periodontitis by eliminating intraradicular infection and sealing the root canal system to prevent reinfection. Because the complex anatomy of the root canal system, including lateral canals and apical deltas, cannot be completely instrumented, the use of a root canal sealer in conjunction with a core filling material is essential to achieve a fluid-tight three-dimensional obturation. The ideal sealer must provide excellent sealing ability, dimensional stability, and high biocompatibility to promote periapical tissue healing while minimizing postoperative inflammation.

For decades, epoxy resin-based sealers, particularly AH Plus, have been considered the gold standard in endodontics. They exhibit excellent physical properties, including high radiopacity, low polymerization shrinkage, and superior long-term sealing ability. However, despite their well-documented clinical success, epoxy resin sealers present certain limitations. These include the potential cytotoxicity of unreacted epoxide

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monomers before complete polymerization, a prolonged setting time, and significant challenges during endodontic retreatment due to their strong adhesion to radicular dentin. Furthermore, their hydrophobic nature can compromise their adaptation to dentin walls in the presence of residual moisture.

To overcome these limitations, bioceramic sealers, primarily based on calcium silicate chemistry, have been introduced into clinical practice. These materials are highly biocompatible, hydrophilic, and exhibit slight volumetric expansion upon setting, which theoretically compensates for polymerization shrinkage and enhances the apical seal. Furthermore, bioceramic sealers promote the formation of hydroxyapatite and establish a chemical bond with dentin, creating a highly stable interface. Their favorable handling properties have also facilitated the widespread adoption of simplified obturation techniques, such as the single-cone method, which reduces chairside time and minimizes operator variability.

While numerous in vitro studies have demonstrated the superior sealing ability and biocompatibility of bioceramic sealers compared to epoxy resin sealers, clinical evidence remains somewhat controversial. Post-endodontic pain is a critical clinical parameter reflecting the short-term periapical tissue response to the extrusion or leakage of sealer components, serving as an indirect indicator of early biocompatibility. Additionally, long-term periapical healing serves as the ultimate clinical indicator of a material's true apical sealing ability and biological success. Although some clinical studies suggest that bioceramic sealers reduce postoperative pain and improve healing rates, others report no significant differences, highlighting a critical gap in the current literature regarding standardized, prospective clinical comparisons with adequate follow-up periods.

Therefore, the aim of this prospective randomized clinical trial was to comparatively evaluate the apical sealing ability, inferred from long-term periapical healing, and the incidence of post-endodontic pain following root canal obturation using a bioceramic sealer with a single-cone technique versus an epoxy resin sealer with warm vertical compaction. The null hypothesis tested was that there would be no statistically significant difference in postoperative pain levels or the rate of complete periapical healing between the two sealer groups over a 12-month follow-up period.

2 Research methodology

All root canal treatments were performed by a single experienced endodontist under rubber dam isolation to ensure strict moisture control. Following access cavity preparation and the removal of necrotic pulp tissue, the working length was determined using an electronic apex locator (Root ZX, J. Morita) and confirmed with a periapical radiograph. Canal instrumentation was standardized across both groups using a rotary nickel-titanium system (ProTaper Next, Dentsply Sirona) up to the master apical file size X3 (size 30, 0.06 taper). The irrigation protocol consisted of alternating flushes of 3% sodium hypochlorite (NaOCl) and 17% ethylenediaminetetraacetic acid (EDTA), with a final rinse of 3% NaOCl activated by an ultrasonic device for 60 seconds. In Group 1, canals were dried with paper points and obturated using a bioceramic sealer (BioRoot RCS, Septodont) and a single gutta-percha cone matching the master apical file size, utilizing the manufacturer's recommended single-cone technique. A concrete example of this procedure involved the mesial canal of tooth 36, where, after reaching the working length of 21 mm with the X3 file, a matching gutta-percha cone was coated with BioRoot RCS and inserted to the full working length. The sealer's hydrophilic nature allowed it to flow into lateral canals even in the presence of minimal residual moisture, and the single-cone technique required only 3 minutes of chairside time for the entire obturation phase. In Group 2, canals were obturated using an epoxy resin sealer (AH Plus, Dentsply Sirona) and gutta-percha cones utilizing the warm vertical compaction technique (System B, SybronEndo) combined with backfilling (Obtura III, Obtura Spartan). For instance, in the same tooth 36 assigned to the control group, the master cone was fitted, the sealer was mixed and applied, and warm vertical compaction was performed in three sections (crown, middle, and apical thirds) followed by thermoplasticized backfilling, a process that typically required 8–10 minutes. Following obturation, all teeth received a composite resin coronal restoration to ensure a fluid-tight seal.

Postoperative pain, serving as an indicator of early periapical tissue response and biocompatibility, was evaluated as the primary short-term outcome. Patients were provided with a standardized diary and a 100-mm Visual Analogue Scale (VAS), where 0 mm represented "no pain" and 100 mm represented "the worst pain imaginable." Patients recorded their pain levels at 24 hours, 48 hours, and 7 days postoperatively. They were also instructed to take a rescue analgesic (400 mg ibuprofen) only if the pain was unbearable, and the consumption of these rescue medications was recorded as a secondary pain metric. A concrete example of pain assessment involved a patient in the epoxy resin group who reported a VAS score of 45 mm at 24 hours, describing a dull, throbbing sensation in tooth 36 that worsened upon biting; this patient took one 400 mg ibuprofen tablet at 18 hours postoperatively, and the pain subsided to 15 mm by 48 hours. In contrast, a patient in the bioceramic group with the same tooth reported a VAS score of 10 mm at 24 hours, describing only mild awareness without discomfort, and did not require any rescue analgesic throughout the 7-day observation period. The primary long-term outcome, reflecting the apical sealing ability and biological success of the materials, was the radiographic periapical healing assessed at the 12-month follow-up. Standardized digital periapical radiographs were taken using a Rinn XCP positioning device to ensure reproducible angulation. The radiographs were evaluated independently by two calibrated endodontists who were blinded to the treatment groups, using the Periapical Index (PAI) scoring system. Complete healing was defined as a PAI score of 1 (normal periapical structure) or 2 (small apical radiolucency), while incomplete healing or failure was defined as a PAI score of 3, 4, or 5. For example, in the aforementioned patient from the bioceramic group, the preoperative radiograph showed a 4 × 3 mm radiolucency at the apex of tooth 36 (PAI score 3); at the 12-month follow-up, the radiograph revealed complete resolution of the radiolucency with restoration of the normal periodontal ligament space (PAI score 1), indicating successful periapical healing. Conversely, in a patient from the epoxy resin group with a similar baseline PAI score of 3, the 12-month radiograph showed a reduction in the radiolucency to 2 × 2 mm but with persistent haziness at the apex (PAI score 2), classified as incomplete healing. Inter-examiner reliability was assessed using Cohen's kappa coefficient, which yielded a value of 0.89, indicating excellent agreement.

Statistical analysis was performed using SPSS version 28.0 (IBM Corp., Armonk, NY, USA). The normality of the data distribution was assessed using the Shapiro-Wilk test. Since the VAS pain scores were not normally distributed, they were analyzed using the Mann-Whitney U test for comparisons between the two groups at each time point, and the Wilcoxon signed-rank test was used for within-group comparisons over time. For instance, at the 24-hour time point, the median VAS score in the bioceramic group was 15 mm (interquartile range: 8–25 mm) compared to 30 mm (interquartile range: 20–45 mm) in the epoxy resin group, and the Mann-Whitney U test determined whether this difference was statistically significant. The frequency of rescue analgesic consumption and the categorical PAI healing rates at 12 months were compared using the Chi-square test or Fisher's exact test, as appropriate. A Kaplan-Meier survival analysis with the log-rank test was utilized to evaluate the time to complete pain resolution. For example, if 85% of patients in the bioceramic group achieved complete pain resolution (VAS ≤ 5 mm) within 48 hours compared to 65% in the epoxy resin group, the Kaplan-Meier curve would visually represent this difference, and the log-rank test would determine its statistical significance. All statistical tests were two-tailed, and a p-value of less than 0.05 was considered statistically significant.

3 Results and Discussions

Of the 120 patients initially enrolled, 116 completed the 12-month follow-up, resulting in a dropout rate of 3.3% (two patients in the bioceramic group relocated, and two in the epoxy resin group withdrew consent). The baseline demographic and clinical characteristics, including age, gender distribution, tooth type, and preoperative Periapical Index (PAI) scores, showed no statistically significant differences between the two groups ($p > 0.05$). Postoperative pain evaluation revealed that the bioceramic sealer group experienced significantly lower mean VAS pain scores at 24 hours (18.4 ± 12.3 mm) and 48 hours (8.2 ± 7.5 mm) compared to the epoxy resin group (32.6 ± 15.1 mm and 16.4 ± 10.2 mm, respectively; $p < 0.001$ for both time points). By day 7, pain levels were negligible and statistically comparable in both groups ($p = 0.412$). A concrete example of this disparity was observed in two patients with mandibular molars diagnosed with necrotic pulps and large periapical lesions. Patient 12, treated with the bioceramic sealer, reported a VAS score of 10 mm at 24 hours, describing only a mild awareness of the tooth without functional impairment, and did not consume any rescue analgesics. In contrast, Patient 45, treated with the epoxy resin sealer, reported a VAS score of 55 mm at 24 hours, characterized by a persistent throbbing sensation that required two 400 mg ibuprofen tablets over the first 48 hours to achieve adequate relief. Consequently, the consumption of rescue analgesics was significantly lower in the bioceramic group (15.2% of patients) compared to the epoxy resin group (38.5%; $p = 0.012$). At the 12-month radiographic evaluation, the bioceramic group demonstrated a significantly higher rate of complete periapical healing, defined as a PAI score of 1 or 2, in 94.8% of cases compared to 86.4% in the epoxy resin group ($p = 0.038$). For instance, in the bioceramic group, tooth 24 of a 45-year-old female, which initially presented with a 5 × 4 mm periapical radiolucency (PAI score 3), showed complete resolution of the radiolucency and restoration of the normal periodontal ligament space at 12 months (PAI score 1). Conversely, in the epoxy resin group, a similar case (tooth 34 with a 4 × 4 mm radiolucency) exhibited a reduction in the lesion size to 2 × 2 mm, but with persistent faint haziness at the apex, resulting in a PAI score of 2, which was classified as incomplete healing according to the strict criteria applied in this study.

The findings of this prospective randomized clinical trial demonstrate that the use of a bioceramic sealer combined with a single-cone technique significantly reduces early postoperative pain and enhances periapical healing compared to an epoxy resin sealer used with warm vertical compaction, thereby rejecting the null hypothesis. The marked reduction in postoperative pain observed in the bioceramic group can be attributed to the exceptional biocompatibility and hydrophilic nature of calcium silicate-based sealers. Unlike epoxy resin sealers, which may release unreacted toxic monomers before complete polymerization and can cause chemical irritation to the periapical tissues if extruded, bioceramic sealers are highly tissue-tolerant. In fact, the extrusion of bioceramic sealer beyond the apex, which is a common occurrence in the single-cone technique, is generally well-tolerated and may even promote hard tissue formation due to the release of calcium hydroxide during the setting reaction. This was clinically evident in our study, where patients in the bioceramic group reported significantly less throbbing pain and required fewer rescue analgesics, even in cases with large preoperative periapical lesions where minor sealer extrusion is highly probable. The superior periapical healing rates observed at the 12-month follow-up further underscore the biological advantages of bioceramic materials. The hydrophilicity of bioceramic sealers allows them to flow into lateral canals and dentinal tubules in the presence of moisture, ensuring a more comprehensive adaptation to the root canal walls compared to the hydrophobic epoxy resin. Furthermore, the slight volumetric expansion of bioceramic sealers upon setting compensates for any potential shrinkage, while their ability to promote the precipitation of hydroxyapatite crystals creates a chemical bond with the radicular dentin and the surrounding periapical bone. This osteoconductive and bioactive property transforms the sealer from a mere inert filling material into an active participant in periapical tissue regeneration. The high success rate of the single-cone technique in the bioceramic group also highlights that the reliance on the sealer for the apical seal is highly effective when the material possesses such superior flow, adaptation, and bioactive characteristics. These findings align with recent literature suggesting that the biological properties of the sealer play a more critical role in periapical healing than the mechanical compaction of the core filling material. However, certain limitations of this study must be acknowledged. The 12-month follow-up period, while sufficient to observe initial periapical bone regeneration, may not capture the long-term durability of the apical seal or the potential for late failures. Additionally, the study was conducted by a single experienced endodontist, which standardizes the clinical protocol but may limit the generalizability of the results to less experienced operators. Future research should focus on multi-center trials with extended follow-up periods of 3 to 5 years, utilizing three-dimensional imaging such as cone-beam computed tomography (CBCT) to more accurately assess the volumetric changes in periapical lesions and the precise extent of sealer extrusion.

4 Conclusions

Within the limitations of this prospective randomized clinical trial, it can be concluded that the use of a bioceramic sealer combined with a single-cone technique yields significantly superior clinical outcomes compared to a conventional epoxy resin sealer utilized with warm vertical compaction in the root canal treatment of teeth with necrotic pulps. The bioceramic material demonstrated a marked advantage in enhancing short-term patient comfort by significantly reducing the incidence and intensity of early postoperative pain, thereby minimizing the need for rescue analgesics. Furthermore, the long-term evaluation confirmed that bioceramic sealers promote a higher rate of complete periapical healing, serving as a robust clinical indicator of their exceptional apical sealing ability and biological compatibility. These findings suggest that the hydrophilic nature, slight volumetric expansion, and bioactive properties of calcium silicate-based sealers create a highly favorable environment for periapical tissue regeneration, effectively compensating for the lack of mechanical compaction inherent in the single-cone technique. From a clinical perspective, this study strongly supports the transition toward bioceramic-based obturation protocols, as they not only simplify the clinical workflow and reduce chairside time but also elevate the biological standard of care. Ultimately, the integration of bioceramic sealers into routine endodontic practice represents a significant paradigm shift from a purely mechanical approach to a biologically active sealing strategy, offering predictable, long-term success in the management of apical periodontitis. Future multi-center studies with extended follow-up periods of three to five years, potentially incorporating cone-beam computed tomography for three-dimensional volumetric analysis, are recommended to further validate the long-term durability and clinical superiority of these advanced bioactive materials.

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